Volume II · Chapter 6 — Transformer Architecture & Self-Attention · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-6-context.md` for the AI interaction log.

Prerequisites: Ch 4 (ML), Ch 5 (NLP basics).

Learning outcomes — the student can: explain self-attention and the transformer block; describe tokenization, embeddings and positional encoding; distinguish encoder/decoder/encoder-decoder families; read a model card and reason about parameters/context length.

Labs (hands-on) — 6 h:

#	Lab	h
6a	Tokenization exploration: tokenize clinical text, inspect subwords & token counts	2
6b	Visualize attention weights on a clinical sentence	2
6c	Run inference with a small pretrained transformer; extract embeddings	2

Datasets/tools: Hugging Face `transformers`; a small BERT/GPT; clinical text samples. **Assessment:** architecture quiz (**50%**); attention-visualization writeup (**30%**); tokenization exercise (**20%**). **Key decisions:** encoder vs. decoder for a task; context length vs. cost; tokenizer choice. **References:** plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory **12** + Lab **6** = **18**.